PFP II: Signed Certificates, Resumable Search, and a Restricted Path-Pairing Obstruction

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Code baseline: c819bec432446dd3c9b1edf1583384cfe76c6c71

Abstract

When a search fails to find a sign transformation, three explanations remain: the transformation does not exist, the search has not finished, or the allowed transformations were too restrictive. This paper gives a small implementation in which those possibilities remain distinct. Within the Proposal Fidelity Protocol (PFP, historically the Symbol Game), a finite signed graph admits either an independently checkable vertex-sign assignment or an odd closed-walk obstruction. An exact real symmetric matrix adapter supplies a complete instance for diagonal sign changes in a fixed basis. A deterministic bounded search records the unexamined suffix and requires replay of every earlier segment before continuation. We also present an exact Fock-space example in which one negative integrated hopping sequence has no positive sequence of equal magnitude. That result rules out a specific pairing operation, not all representations of the Hamiltonian. The contribution is a reproducible certificate and evidence protocol joining established signed-graph reasoning to honest incomplete search, with a worked physical example whose representation boundary is explicit.

1. The question must include the presentation

Imagine changing the signs of rows and the corresponding columns of a symmetric matrix. Can every nonzero off-diagonal entry become negative? This is a precise finite question. It differs from asking whether an arbitrary change of basis, a determinant identity, or a regrouping of paths can produce a useful nonnegative simulation.

A *presentation* records the object, its representation, and the permitted moves. A negative answer concerns that entire tuple. PFP stores this tuple beside a candidate and asks the host to verify the candidate again; a retrieved answer is a proposal until checked.

The running matrix is

$$H = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}.$$

The diagonal matrix $D = \operatorname{diag}(1, -1, 1)$ turns all three off-diagonal pairs negative in DHD. A reader can verify this by multiplying each entry H_{ij} by $D_{ii}D_{jj}$. Conversely, a triangle whose three off-diagonal pairs are all positive cannot be repaired by these sign changes: each of its three edges asks its endpoints to receive opposite signs.

These examples are small enough to inspect completely. Their value is the clean boundary between an answer, a witness, and the larger scientific question that motivated the matrix.

2. Signed graphs and a complete certificate pair

Let V be a finite ordered list of distinct vertex names and E a finite list of undirected edges. Multiple edges and self-loops are allowed. Give each edge $e = uv$ a bit $c_e \in \{0, 1\}$. We seek bits α_v satisfying

$$\alpha_u \oplus \alpha_v = c_e \quad \text{for every edge } e.$$

Here $\oplus$ is addition modulo two. An edge with bit one requires opposite vertex bits; an edge with bit zero requires equal bits. Equivalently, switching vertex signs should turn every edge sign $(-1)^{c_e}$ positive.

Signed-graph criterion. Such an assignment exists if and only if every closed walk has total edge parity zero.

For necessity, XOR all edge equations around a closed walk. Every vertex occurrence contributes twice, so the left side vanishes. For sufficiency, choose a root in each component and set its bit to zero. Assign every other vertex the XOR of edge bits along a path from the root. Two choices of path form a closed walk, so the assumed condition makes this assignment independent of the choice. Appending one edge establishes the required endpoint equation.

The proof yields two independently checkable witness types:

Outcome	Witness	Host checks
A gauge exists	One bit for every vertex in the declared order	Every endpoint equation
No gauge exists in this grammar	A nonempty closed walk with odd total parity	Edge identities, adjacency, closure, and XOR

The implementation in sign_graph.py constructs a spanning forest by breadth-first traversal. A conflicting edge closes an odd walk through the tree. If there is no conflict, the assigned bits form a gauge. Construction is linear in the graph's vertices and edges. The separate witness verifiers do not trust the constructor's verdict.

For the running example, the edge bits on $01, 12, 20$ are $1, 1, 0$. The assignment $(0, 1, 0)$ works. For the all-positive matrix, the edge bits are $1, 1, 1$, and the triangle itself witnesses parity one. A single odd self-loop is the smallest obstruction because it requires $\alpha_v \oplus \alpha_v = 1$.

This is the classical balance criterion for signed graphs (Harary, 1953), presented here as a host-checkable protocol. We claim no new graph theorem.

The parity carrier must not be confused with the rational edge reports of the companion paper PFP I (Carroll, 2026; see docs/references.md). Reports $(1, 1, 0)$ around an oriented triangle have even parity but rational sum two. Their uniform-weight rational repair cost is two. The coefficient system and the allowed changes are part of the claim.

3. Exact matrices and the boundary of stoquasticity

Assume H is a finite real symmetric matrix with entries represented by integers or exact fractions in a declared basis. For every nonzero off-diagonal entry, assign

$$c_{ij} = \mathbf{1}[H_{ij} > 0], \qquad D_{\alpha,ii} = (-1)^{\alpha_i}.$$

Then

$$(D_\alpha H D_\alpha)_{ij} = (-1)^{\alpha_i + \alpha_j} H_{ij}.$$

Therefore all its nonzero off-diagonal entries are negative exactly when $\alpha_i \oplus \alpha_j = c_{ij}$. Zero entries impose no equation. The signed-graph checker completely decides this fixed-basis diagonal-sign problem. It rejects floating-point, nonsquare, and nonsymmetric inputs rather than silently changing the mathematical question.

A real symmetric matrix with nonpositive off-diagonal entries is called *stoquastic* in that basis. If H' has this property and $\beta \geq 0$, choose $a \geq \max_i H'_{ii}$. Then $B = aI - H'$ is entrywise nonnegative, and

$$e^{-\beta H'} = e^{-\beta a} \sum_{k=0}^{\infty} \frac{\beta^k B^k}{k!}$$

is entrywise nonnegative. Finite dimensionality ensures convergence under declared assumptions, and each summand has nonnegative entries. This establishes a positivity statement for this matrix expansion; it does not establish rapid sampling or a low-variance estimator for every observable.

Other transformation grammars can answer different questions. Allowing an unrestricted global eigenbasis makes any finite Hermitian matrix diagonal, but obtaining that basis and expressing observables may be computationally prohibitive. Local basis changes are more constrained and more general than our diagonal signs. Klassen and Terhal (2019) analyze such changes for two-local qubit Hamiltonians; their results are not supplied by our graph checker.

4. Search that can stop without inventing an obstruction

The complete graph algorithm is already efficient. Why retain an exponential candidate search? Its purpose is to exercise the protocol used when a future candidate grammar lacks such a simple complete solver.

frontier_search.py enumerates the $2^{|V|}$ bit assignments in lexicographic order. It checks fixed batches of at most eight, independent of worker count, and consumes completed results in canonical order. Thread scheduling therefore cannot select a different first witness or reorder the recorded checks. This is deterministic batch enumeration; it has no Monte Carlo tree-search policy and makes no CPU speedup claim.

A receipt records the presentation and graph digests, start index, budget, ordered candidate checks, remaining suffix, witness if present, and a digest of that record. Its statuses are:

- **GAUGE_VERIFIED:** a returned gauge passes the exact verifier.
- **UNKNOWN:** the tested segment failed and an unexamined suffix remains.
- **RESTRICTED_OBSTRUCTION:** the finite grammar is exhausted and the independent graph constructor supplies a verified odd walk.

In the running three-vertex example, indices zero and one represent 000 and 001, both invalid. With a budget of two, the correct outcome is UNKNOWN and the remaining interval is $[2, 8)$. Resuming with a budget of one examines 010, verifies it, and leaves the cursor at three. This is a worked control trace, not a benchmark of search efficiency.

A nonzero starting cursor is accepted only when prior UNKNOWN receipts replay as a contiguous chain beginning at zero. Each recorded bit vector and verdict is checked again; a recomputed digest by itself is insufficient. Skipping an earlier segment or importing a chain for a different graph must fail. Completeness is therefore tied to verified coverage, rather than to a caller's assertion that earlier work happened.

The presentation key includes Hamiltonian digest, basis, ensemble, filling, boundary, representation, estimator, observables, inverse temperature, parameters, transformation grammar, adapter digest, and checker version. The index intersects all key fields and then tests full-key equality. Even identical Hamiltonian bytes do not justify moving a gauge between different declared representations. Matching a key still does not prove that the claimed Hamiltonian-to-matrix adapter is scientifically correct.

5. What an equal-weight pairing must prove

For a finite signed sum

$$Z = \sum_{x \in \Omega} \sigma(x) A(x), \qquad \sigma(x) \in \{-1, 1\}, \quad A(x) > 0,$$

a cancellation certificate can specify a subset $P \subseteq \Omega$ and a map $I : P \to P$. It must verify $I(I(x)) = x$, $I(x) \neq x$, $\sigma(I(x)) = -\sigma(x)$, and $A(I(x)) = A(x)$. The paired contribution then cancels term by term. If all survivors have nonnegative contribution, the remaining sum is nonnegative.

Checking only that the total is nonnegative is weaker. Contributions $+2, -1, -1$ sum to zero, but no negative term has a positive term of equal magnitude. Pairing every term of a positive thermal partition function is also obstructed: for a Hermitian Hamiltonian on a nonzero finite-dimensional Hilbert space and real finite β, $\operatorname{Tr} e^{-\beta H} > 0$. A complete cancellation would give zero. The same argument applies to an absolutely convergent expansion when all terms are paired.

These observations motivate a sharply bounded physical example. Its certificate excludes one operation on one carrier; it neither classifies all algorithms nor rules out regrouping several terms together.

6. A restricted Fock-space obstruction with exact bounds

Use a three-site triangle with spin-up modes followed by spin-down modes in the order

$$(\uparrow 0, \uparrow 1, \uparrow 2, \downarrow 0, \downarrow 1, \downarrow 2).$$

An occupation state is a six-bit string, so there are $2^6 = 64$ states. Set hopping amplitude $t = 1$, on-site repulsion $U = 2$, chemical potential zero, and inverse temperature $\beta = 1/4$. The diagonal energy is

$$E(x) = 2\sum_{i=0}^{2} n_{\uparrow i} n_{\downarrow i}.$$

The hopping matrix element of $-H_{\mathrm{hop}}$ moves one particle between two different sites of the same spin and is $(-1)^q$, where q counts occupied modes strictly between the two positions in the declared mode order.

Our carrier consists of rooted closed hopping sequences after their ordered jump times have been integrated out. For a sequence with n jumps, its weight is the product of hopping signs times

$$\int_{0<\tau_1<\cdots<\tau_n<\beta} \exp\left[-\sum_{j=0}^{n} E(x_j)(\tau_{j+1}-\tau_j)\right] d\tau_1 \cdots d\tau_n,$$

where $x_n = x_0$, $\tau_0 = 0$, and $\tau_{n+1} = \beta$. Sequences at different lengths belong to this carrier; it is not a finite set solely because its state space is finite.

Consider the three-hop cycle

$$110;111 \longrightarrow 101;111 \longrightarrow 011;111 \longrightarrow 110;111.$$

Its hopping-sign product is negative, every visited state has energy four, and its integrated weight is exactly

$$-W_\star = -e^{-1}\frac{(1/4)^3}{3!} = -\frac{e^{-1}}{384}.$$

The retained exact checker enumerates all rooted three-hop sequences. Its asserted finite facts are 192 sequences, evenly split into 96 positive and 96 negative, with every positive three-hop sequence visiting only energies at most two. It also checks the global energy range $0 \le E \le 6$, absence of one-hop closures, and symmetric unit-magnitude hopping.

These facts imply the following bounds for an individual positive sequence:

Jump count	Positive integrated weight
0	At least $e^{-3/2}$
1	No closed sequence
2	At least $e^{-3/2}/32$
3	At least $e^{-1/2}/384$
$n \ge 4$	At most $(1/4)^n/n! \le 1/6144$

The lower bounds use the maximum energy during the path and the simplex volume $\beta^n/n!$. The upper bound uses nonnegative energies and unit-magnitude hopping. Its final expression decreases for $n \ge 4$. It bounds one sequence, not the sum of all long sequences.

The strict separations

$$\frac{1}{6144} < W_\star < \frac{e^{-1/2}}{384}, \qquad W_\star < \frac{e^{-3/2}}{32} < e^{-3/2}$$

exclude every possible positive partner on the declared carrier. The checker establishes these inequalities with rational enclosures for e^{-1}, $e^{-1/2}$, and $e^{-3/4}$, squaring the last lower bound. For $0 \le x \le 1$, the alternating series for e^{-x} has decreasing term magnitudes; odd partial sums are lower bounds and even partial sums are upper bounds. Thus no floating-point comparison is needed.

The conclusion combines finite enumeration at length three with an analytic bound covering every longer length. A sign-reversing equal-magnitude map that covers this negative sequence does not exist on this carrier. Time-labelled maps before integration, determinant identities, collective regroupings, other bases, and other estimators remain outside the statement. Establishing a useful sign-free simulation in any such representation requires its own construction and observable-preservation argument.

7. Reproduction, custody, and negative controls

The four frozen finite-graph files were copied from the author's non-Git Genesis workspace on 2026-09-23. The matrix snapshot repair of 2026-09-24 changed the search module and one regression test. The provenance record retains both original and repaired hashes and records later packaging changes with their current hashes. The baseline named above supplies the exact repository revision for the sources discussed here.

The repaired code snapshots matrix rows before validation, hashing, and classification. Before that repair, a one-shot iterator could be consumed during validation and subsequently interpreted as an empty matrix. This is a concrete reason to bind checking and hashing to the same immutable input.

From the package root, run one test file per command. The commands use `py -3.12`, the Windows launcher for Python 3.12; on other systems, substitute a Python 3.12 interpreter.

```
py -3.12 -m pytest code/tests/test_sign_graph.py -q
py -3.12 -m pytest code/tests/test_frontier_search.py -q
py -3.12 -m pytest code/tests/test_manuscript_examples.py -q
py -3.12 -m pytest code/tests/test_exact_fock_gap.py -q
py -3.12 -m pytest code/tests/test_native_sign_graph.py -q
py -3.12 -m pytest code/tests/test_native_refusals.py -q
py -3.12 code/exact_fock_gap.py
py -3.12 code/run_demos.py --all
```

The file test_manuscript_examples.py checks the worked examples of Sections 1, 2, and 4. The exact checker prints the 192, 96, and 96 path counts and the rational bounds of Section 6. The dated review receipt for this revision records the interpreter, each command, tool versions, exit statuses, and the tested scope; see "Verification for this revision". Historical 18-test and 19-test receipts apply to their dated snapshots only. The package also contains an optional native Rust implementation of the graph checker, described in the README; the statements of this paper rest on the proofs and the Python checkers.

Useful negative controls include an odd self-loop, an all-positive triangle, a malformed closed walk, floating-point matrix input, a changed presentation key, a skipped frontier interval, and a one-shot matrix iterator. The notebook illustrates the finite checker; its saved outputs are not independent experimental data.

This package contains no machine-checked proof of the signed-graph implementation. The small formal proofs accompanying the companion paper PFP I, a pair-cancellation law and a triangle sign obstruction, concern their own restricted types. Historical model conversations and their terminal labels are also insufficient to establish physical preservation, model success rates, or current runtime operation.

8. Related work and practical use

Harary's signed-graph balance result supplies the underlying parity criterion. Our contribution is the explicit witness interface, the bounded search record, and the separation of complete graph classification from incomplete exploration. The classical graph algorithm is the right baseline when this exact grammar is the entire problem.

Troyer and Wiese (2005) establish worst-case computational difficulty for a generic solution of the fermion sign problem. That result does not certify failure of an individual Hamiltonian, basis, or pairing proposal. Our Fock example provides a different kind of evidence: a concrete obstruction to a specifically defined operation.

The methods are useful for testing certificate interfaces, replaying bounded search, and identifying where an adapter must supply new scientific evidence. A physical application must pin its Hamiltonian, ensemble, basis, boundaries, mode ordering, measure, observables, and transformation grammar. It must show preservation of the partition function and observable numerators, then analyze estimator cost if computational usefulness is claimed. This package implements finite graph and matrix checks and the restricted path example. It does not provide that general physical adapter.

Verification for this revision

The package verifier (`verify_deposit.py`) checks the manifest and the two-way file census; `tools/rebuild_release.py --check` checks the release archive. The package tests, lint, and notebook replay are executed by the dated review receipt recorded with this revision; that receipt lists each command, tool version, exit status, and tested scope. Historical receipts in this package keep their original dates and scope and are not current execution evidence.

References

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- Harary, Frank (1953). "On the notion of balance of a signed graph." *Michigan Mathematical Journal*, 2(2), 143–146. doi:10.1307/mmj/1028989917.
- Klassen, Joel, and Barbara M. Terhal (2019). "Two-local qubit Hamiltonians: when are they stoquastic?" *Quantum*, 3, article 139. doi:10.22331/q-2019-05-06-139; author preprint, arXiv:1806.05405.
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